MIL-C-21522 (Wep)
16 February1960

MILITARY SPECIFICATION

CYLINDERS , LOCKOUT FOR SHOCK ABSORBER, MISSILE LAUNCHER MARK 2 MOD 0

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification states the requirements of two types of lockout cylinders for-use in a shock absorbing system.

1.2 Classification - Lockout cylinders shall be of the fallowing types:

Type I Horizontal double acting

Type II Vertical, single acting

2. APPLICABLE DOCUMENTS

2.1 Documents - The following documents, to the extent stated herein, of the issue in effect on the date of the invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-S-901	Shockproof Equipment, Class HI (High Impact), Shipboard Application, Tests for
MIL-E-5272	Environmental Testing, Aeronautical-and Associated Equipment, General Specifications for

FSC 1440

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment.

DRAWINGS

NAVY DEPARTMENT

BuOrd LD 419531	Cylinder, Lockout, Vertical, Single-Acting, Assembly
BuOrd LD 419548	Cylinder, Lockout, Horizontal, Double-Acting, Assembly
SA-2440175	Operational Test Setup for Liquid Springs
SA-2440366	Pneumatic Hi-Pressure Test
SA-2440493	Vibration Test Fixture for Lockout Cylinder and Liquid Springs
SA-2619427	Shock Test Fixture
SA-2619682	Support System Campo-nent Fixture

PUBLICATIONS

NAVY DEPARTMENT

OP 400	General Instructions for Design, Manufacture, and Inspection of Naval Ordnance Equipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.

NAVORD INST 4355.1	Equipment for Use by Government Inspectors in Final Inspection of Naval Ordnance Items
NAVORD INST 4355.28	Bureau of Ordnance Test Calibration Program
Standards Laboratory Information Manual	

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other Publications - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 and all supplements thereto	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents. Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station., Chicago 6. Illinois.)

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description - The lockout cylinder consists of a cylinder body, piston rod, and, according to type, one or two free pistons. Normally, the cylinder is not energized and the piston rod is free to move axially within limits. During operation, each lockout cylinder is energized with high-pressure air. In the horizontal cylinder, when pressurized, the two free pistons move to a predetermined central location, bringing the piston rod to a median position where it resists displacement in either directions In the vertical cylinder, when pressurized, the single free piston moves to a predetermined central location, moving the piston rod, if extended, to the median position where it resists displacement in extension only.

3.2 General Requirements

3.2.1 Preproduction Samples - Unless otherwise specified in contract or order, preproduction samples of the lockout cylinders shall be manufactured using the methods and procedures proposed for the production lots These samples will be tested as specified in section 4 herein, and are for the purpose of determining that the production item meets the requirements of this specification (refer to paragraph 4.2.1). At the discretion of the procuring agency, the requirement for a preproduction sample may be waived and shall then be satisfied by substitution of a production unit chosen at random by the Government Inspector.

3.2.2 Materials - Unless otherwise specified in contract, requisition, or order, all materials and purchased parts entering into the final assembly of the lockout cylinders shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BuOrd LD 419531 and 419548.

3.2.3 Finishes, Coatings, Construction and Dimensions - Finishes, coatings, construction, and dimensions for the lockout cylinders shall be in accordance with the applicable drawings, specifications, and publications, listed in BuOrd ID 419531 and 419548.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.

3.2.4 Marking - Marking of assemblies, subassemblies, and parts shall be in accordance with applicable drawings listed in BuOrd LD 419531 and 419548.

3.2.5 Interchangeability - Individual assemblies and/or piece parts of assemblies fabricated in agreement with their applicable drawings and specifications shall be interchangeable with their like numbered parts.

3.2.6 Gages - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies, will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, requisition, or order, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Threads - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28 and all supplements thereto. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Request for Deviation - The procedures for requesting deviations from this specification, or any specifications publications, materials, or processes specified herein, shall be as set forth in OP 400.

3.2.9 Conflicting Requirements - In event of conflict between this specification and any specification, publication, or drawing referenced herein, this specification shall govern.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

3.3 Performance Requirements

3.3.1 Performance

3.3.1.1 Vertical Lockout Cylinder - The vertical lockout cylinder, when energized with an air pressure of 4,000 psig, shall return the piston rod from the extreme extended position to the median position and while at the median position shall resist, without displacement, a force of 28,700 pounds acting to extend the piston rod, spall show no damage, and, there after shall meet all performance requirements and product characteristics of this specification.

3.3.1.2 Horizontal Lockout Cylinder - The horizontal lockout cylinder, when energized with an air pressure of 4,000 psig, shall return the piston rod from the extreme travel positions and while at the median position shall resist, without displacement, a force of 11,000 pounds acting to move the piston rod in either direction, shall show no damage, and thereafter shall meet the performance requirements and product characteristics of this specification.

3.3.2 Hydrostatic Pressure

3.2.l Lockout Cylinder Body - The lockout cylinder body (before assembly) shall withstand a hydrostatic proof pressure of 7,500 psig for 10 minutes and shall show no leakage or damage.

3.3.2.2 Lockout Cylinder - The lockout cylinder, assembled, shall withstand a hydrostatic proof pressure of 7,500 psig for 30 minutes, shall show no damage, and thereafter shall meet all the performance requirement and product characteristics of this specification.

3.3.3 Leakage - After being energized twice by air at 4,000 psig, the lockout cylinder, when actuated by air at 4,000 psig and allowed to stand for 1 hour with no replenishment, shall show no pressure loss in excess of 200 psig from the 4,000 psig pressure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.

3.3.4 Operating Life - The lockout cylinder shall have an operating life of 1,000 cycles, with no maintenance, at a maximum piston-rod travel rate of 68 inches-per-minute. As the piston rod reaches median position in the cycle, each piston shall be returned from the extreme travel position to the median position by an air pressure of 4,000 psig. For the purpose of this requirement, a cycle shall be defined as follows:

Vertical Cylinders:	From median position, to extension of 5-5/8 ± 1/8 inches, to median position, to compression of 5-5/8 ± 1/8 inches, and back to median.
Horizotal Cylinders:	From median position, to extension of 3-5/8 ± 1/8 inches, to median position, to compression of 3-5/8 ± 1/8 inches, and back to median.

After Cycling, the lockout cylinders may show signs of wear and damage but shall meet all performance requirements and product characteristics of this specification.

3.4 Environmental Conditions

3.4.1 Vibration., Installed Condition - The lockout cylinders, unpressurized, shall withstand the steady-state vibration, in accordance with Type I, Standard MIL-STD-167, in the range of 5 to 25 cycles-per-second, shall show no damage, and thereafter shall meet all the performance requirements and product characteristics of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.

3.4.2 Vibration, Shipping Condition - The lockout cylinders, packaged and packed in accordance with section 5 herein, shall withstand the following vibration:

VIBRATION CRITERIA (MAX) FOR PACKAGING

Type of Transportation	Vertical cps	Vertical g	Longitudinal cps	Longitudinal g	Lateral cps	Lateral g
Truck	200	2.5	150	1.5	200	1.0
	10	1.0	10	1.0	10	1.0
Rail	60	2.0	60	1.0	30	1.0
	10	1.0	10	0.5	10	0.5
Aircraft	60-100	1.0	60-100	1.0	60	1.0
	100-200	0.5			100	0.5
Ship (Surface)	10	0.5	Negligible		10	0.25
	30	1.5			30	0.75

3.4.3 Shock, Installed Condition - The lockout cylinders, unpressurized shall withstand the shock test in accordance with Type A, Specification MIL-S-901, for Grade 1 equipment. With the piston-rod eye pinned to a stationary clevis external to the shock-test machine, shock loading shall be applied at the lockout-cylinder-body eye. The test The test fixture shall be so designed that the piston-rod travel shall not exceed the excursion limits specified in 3.3.4. After testing, the lockout cylinders shall show no damage and shall meet all performance requirements and product characteristics of this specification

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

4.3 Classification of Tests - The inspection and testing of the lockout cylinders shall be classified as follows:

4.3.1 Production Tests - These tests are to be accomplished on each lockout cylinder being submitted for acceptance under the contract. Production tests shall be either performed by the manufacturer and witnessed by the Government Inspector, or the tests may be conducted by the Government Inspector. These tests are detailed in 4.5.

4.3.2 Preproduction Tests - Preproduction tests are those which are accomplished on samples selected as specified in 4.2.1, which are representative of the Production of the lockout cylinders, to determine that the Production meets the requirement of this specification. These are detailed in 4.5 and 4.6. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the cessation of production until determination of cause of failure is made and appropriate correction approved by the procuring activity.

4.3.3 Tests After Delivery - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein as deemed necessary after installation of the lockout cylinder. Any defect discovered before, during, or after these tests, due to the contractor's failure to comply with the drawings or specifications for this equipment, shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.

4.4 Special Test Equipment - The following items of special test equipment are required to perform the acceptance tests set-forth in this specification.

SA-2440175	Operational Test Setup for Liquid Springs
SA-2440366	Pneumatic High Pressure Test
SA-2440493	Vibration Test Fixture for Lockout Cylinder and Liquid Springs
SA-2619427	Shock Test Fixture
SA-2619682	Support System Component Fixture

4.5 Production Tests

4.5.1 Responsibility - The contractor or Government Inspector shall conduct production tests as specified in 4.3.1 to assure that the lockout cylinder is in compliance wiith the requirements of 3.3.1 through 3.3.3.

4.5.2 Test Conditions - Unless otherwise specified the lockout cylinders shall be subjected to production tests under the following conditions:

Temperature:	Room ambient, +65 to +100 degrees Fahrenheit
Pressure:	14.7 ± psia
Vibration:	Not to exceed 1/4 g acceleration
Humidity:	Relative humidity of 100 percent or less

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.

4.5.3 Test and Inspection Equipment and Facilities- The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all production tests. The test equipment shall be adequate in quantity and, unless definite requirement are specified, shall be of sufficient accuracy and quality to permit performance of the required production tests,

4.5.3.1 Calibration - Measuring equipment and instrumentation shall be calibrated as required by NAVORD INST 4355.1, 4355.28, and in accordance with the policies specified in the Standards Laboratory Information Manual issued by the Bureau of Naval Weapons Representative, Pomona, California.

4.5.4 Test Procedure

4.5.4.1 Materials and Dimensions - Each lockout cylinder shall be inspected for conformance to the requirement Of 3.2.2.

4.5.4.2 Hydrostatic Test - Each lockout cylinder body (unassembled) shall be subjected to a hydrostatic proof pressure of 7,500 psig for 10 minutes to meet the requirements of 3.3.2.

4.5.4.3 Leakage Test - Each lockout cylinder, after being energized twice, when charged with an air pressure of 4,000 psig and allowed to stand for 1 hour, shall meet the requirements of 3.3.3.

4.5.4.4 Performance Test

4.504.4.1 Vertical Lockout Cylinder - Each vertical lockout cylinder, energized with an air pressure of 4,000 psig and subjected to a force of 28,700 pounds acting to extend the piston rod from the median positions shall meet the requirements of 3.3.1.1.

4.5.4.4.2 Horizontal Lockout Cylinder - Each horizontal lockout cylinder, energized with an air pressure of 4,000 psig and subjected to a force of 11,OOO pounds acting to move the piston rod from the median position, shall meet the requirements Of 3.3.1.2.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

4.6 Preproduction Tests

4.6.1 Vibration Test, Installed Condition - One lockout cylinder of each type, unpressurized and suitably mounted in a test fixture, shall be subjected to vibration tests in accordance with standard MIL-STD-167 to meet the requirements of 3.4.1.

4.6.2 Vibration Test, Shipping Condition - One lockout cylinder of each type, packaged and packed as specified in section 5 herein, and suitably mounted in a test fixture, shall be subjected to vibration tests to meet the requirements of 3.4.2.

4.6.3 Shock Test, Installed Condition - One lock-out cylinder of each type, unpressurized, shall be subjected to shock tests to meet the requirements of 3.4.3.

4.6.4 Hydrostatic Test - One lockout cylinder of each type shall be subjected to a hydrostatic test to meet the requirements of 3.3.2.2.

4.6.5 Corrosive Atmosphere Test - One lockout cylinder of each type shall be exposed to the salt spray test, in accordance with Specification MIL-E-5272, to meet the requirements of 3.4.5.

4.6.6 Operating Life Test - One lockout cylinder of each type shall be subjected to 1,000 cycles of operation to meet the requirements of 3.3.4.

4.6.7 Rejection - Failuure of the lockout cylinders-to meet any of the requirements and tests of this specification shall be considered cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5 1 Preservation and Packaging - Preservation and packaging of the lockout cylinders shall be Level A or Level C.

5.1.1 Level A - Level A shall be as specified by the procuring activity.

5.1.2 Level C - Cleaning, drying, preservation and unit packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing - Packing of the lockout cylinders shall be Level A or Level C.

5.2.1 Level A - Level A shall be as specified by the procuring activity.

5.2.2 Level C - The lockout cylinders, packaged as specified in 5.1.2, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall comply with Uniform Freight Classification Rules or other common-carrier regulations applicable to the mode of transportation.

5.3 Marking - In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Ordering Data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of preservatian, packaging, and packing.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Govornent may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permision to manufacture, use, or sell any patented inventlon$_a$ that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T09:47Z
Check the source to verify that this is the currenversion before use.